

Comprehensive Analysis and Dietary Strategies with Tableau

A College Food Choices Case Study • Folder 2: Requirement Analysis Phase

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Folder 2 Metadata & Requirements Profile

System Registry Parameter	Value Specification
Primary Deliverable	Technical Requirements Specifications Document (TRD)
Data Ingestion Layout	Unstructured Questionnaire CSV -> Tableau Engine Store (.hyper Extract)
Operational Objectives	Latency Mapping, Column Variable Typing, Metric Calculations

1. Technical Requirements Executive Summary

Following the initial structuring phase, Folder 2 translates conceptual user pain points directly into software engineering criteria. This documentation sets up the exact database field boundaries, system dependencies, and engine constraints required to maintain low dashboard loading latency and error-free rendering.

2. Data Schema & Variable Mapping

The tracking schema breaks down survey questionnaire items into precise categorical classes inside the metadata layer:

Field Name Source	Data Type	Role Mapping	Description & Range Scale Limits
GPA	Decimal (Float)	Continuous Measure	Student's cumulative GPA matrix value. Permitted range: 0.0 - 4.0.
Stress_Level	Integer (Discrete)	Discrete Dimension	Self-reported tracking scale value ranging from 1 to 10.
Financial_Status	Integer (Discrete)	Discrete Dimension	Categorizes student weekly budget parameters (1 = High, 5 = Low).
Exercise_Frequency	Integer (Discrete)	Discrete Dimension	Weekly workout indicators (1 = Daily, 2 = 3-4 Days, 3 = None).

3. Functional Requirements Matrix (FRM)

Req ID	Function Name	Technical Target Specifications
FR-01	Multi-Variable Filters	Tableau charts must automatically recompute values when adjusted by gender or GPA filters.
FR-02	Stress Diet Correlation	Calculates and maps calorie, fat, and sugar averages across specific stress index values.
FR-03	Budget Performance Card	Groups diet scores dynamically into low vs high student spending capability categories.

4. Non-Functional Requirements Matrix (NFR)

Req ID	NFR Category	Performance Target Benchmarks
NFR-01	Engine Processing Speed	All dashboard calculations and action filter triggers must re-render in under 2.5 seconds.
NFR-02	Visual Layout Design	Interface placement must align with F-shape viewing grids using colorblind-safe color contrasts.
NFR-03	Data Sanitization Guard	Null inputs or invalid outliers must be filtered out during preprocessing before database loading.

5. Hardware, Software, and System Environment Specs

Environment Layer	Minimal Operational Specifications
Workstation Hardware	Intel i5 / AMD Ryzen 5 CPU, Minimum 8 GB RAM (16 GB optimal for hyper cache).
Software Stack	Windows 10/11 x64, Tableau Desktop Pro v2024.1+, Excel Data Engine.
Production Server	Tableau Public Cloud Platform hosting environment.

6. Key Performance Indicators (KPI) Glossary

KPI Target Metric	Mathematical Calculation Logic	Strategic Value
Dietary Quality Index (DQI)	Daily whole food servings divided by daily processed sugar inputs.	Tracks general health values across chosen demographics.
Stress Comfort Coefficient	Pearson r correlation value calculated between continuous stress values and snack intake.	Identifies academic anxiety food triggers early.

7. Folder 2 Conclusion & Next Design Steps

The creation of this Technical Requirements document secures a precise specification baseline. By mapping dataset variables, system boundaries, and processing constraints clearly, the document forms an actionable technical blueprint for moving safely into Folder 3: Project Design.

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Review Matrix:

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